

NURSES CALLING SYSTEM

Revision: 0.002

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<http://20.163.9.187:3000/> - USER IP

<http://20.163.9.187:5001/> - DEVICE IP

EMAIL : vishnu@example.com

PASSWORD: [Admin](#)

Project Overview:

The **Nurses Calling System** is a communication system used in hospitals that allows patients to call nurses quickly when they need assistance. The system improves patient care by providing an efficient way for patients to request help without leaving their bed.

The system usually consists of **call buttons placed near patient beds**, a **central monitoring unit**, and **notification systems** such as lights, buzzers, or display panels.

Objective:

- To provide **quick communication** between patients and nurses.
- To improve **patient safety and response time**.
- To reduce **manual monitoring workload** for nurses.
- To provide **real-time alerts** when patients require help.

Scope of the Project:

The Nurses Calling System can be used in:

- Hospitals
- Clinics
- Elderly care homes
- Rehabilitation centers
- ICU wards

The system can be implemented using **embedded systems, IoT, or software-based monitoring systems**.

System Overview:

The system works when a patient presses a **call button**. The signal is sent to a **central controller or server**, which then alerts the **nursing station** through:

- Visual indicators
- Alarm sounds
- Display panels
- Mobile notifications

System Architecture:

1. Components of the system:
2. Patient Call Unit
3. Central Controller
4. Nurse Station Display
5. Alert System
6. Database (optional for logging)

Patient Button → Microcontroller → Server → Nurse Station Display



Database

Software Requirements:

Software used for development:

- Backend: Node.js
- Database: SQL Server
- Frontend: React.js

Functional Modules:

1. Patient Module:

- Patient presses call button
- Signal sent to system
- Request is logged

2. Nurse Station Module:

- Displays patient room number
- Shows active calls
- Allows nurse to acknowledge the request

3. Alert System:

- Sound alarm
- LED indicator
- Display message

4. Admin Module

- Manage users and rooms
- Monitor call history
- Generate reports

System Workflow:

- Patient presses the **call button**
- Signal is sent to the **controller**
- System identifies **patient room**
- Notification is sent to **nurse station**
- Nurse acknowledges the call
- System updates the **call status**

Advantages:

- Faster response time
- Improved patient care
- Reduced nurse workload
- Real-time monitoring
- Emergency alerts

DETAILS:

OID= ORG001 (organization id)

SOLAR MONITOR string:

http://20.163.9.187:5001/api/callstatus/insert
orgId=ORG001&dnum=101&floor=0&status=1

Send data --> Service --> Database --> UI

PARAMETER DETAILS:

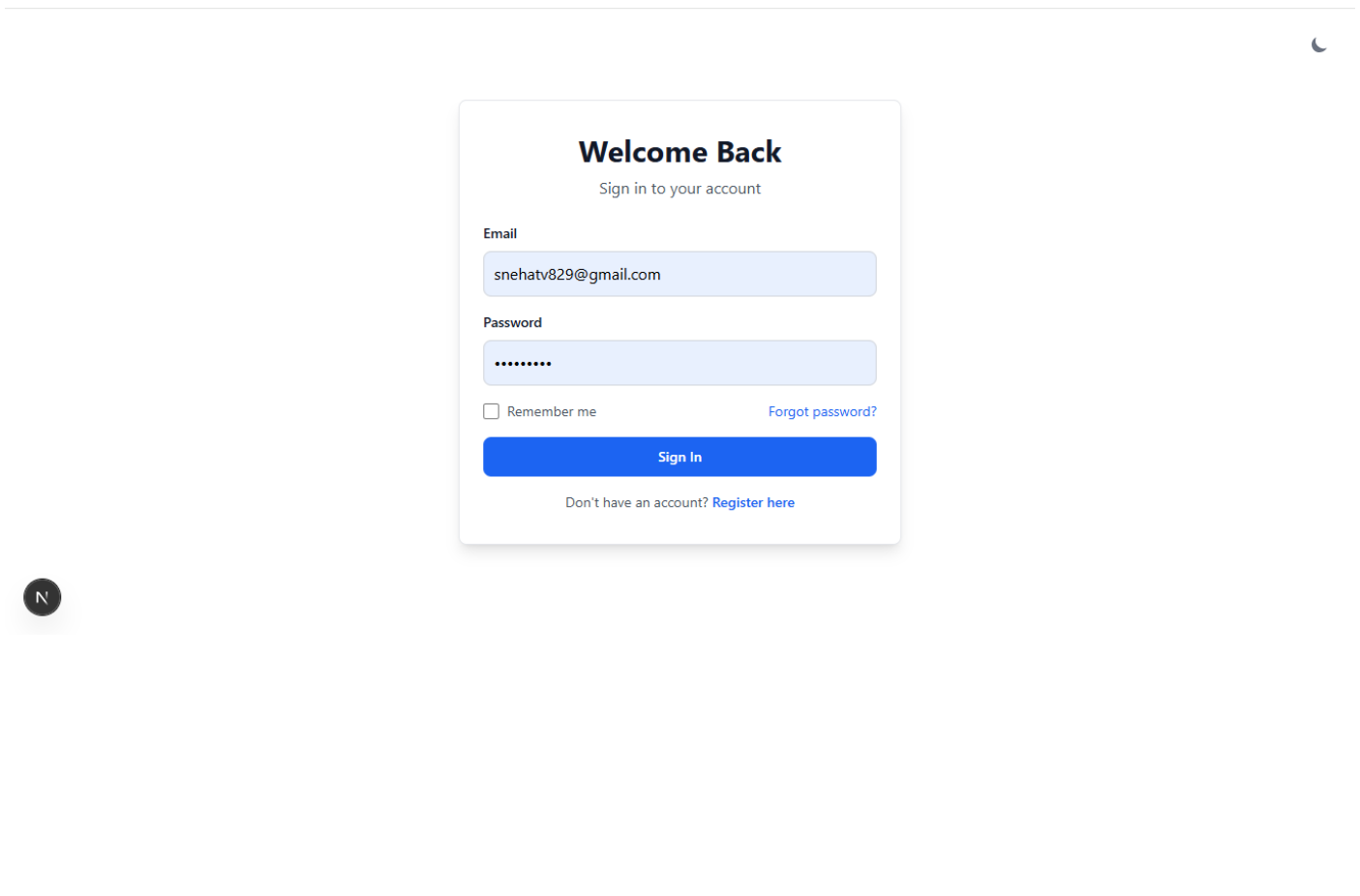
dnum	ROOM NO.
floor	FLOOR NO.
status	STATUS => 1-CALL, 0-MUTE, 2-RESET

Future Enhancements:

- Mobile app notifications for nurses
- Integration with hospital management system
- Voice communication
- AI-based emergency detection
- Wireless IoT sensors

SCREENSHOTS:

LOGIN PAGE:



DASHBOARD:

← → ↻

Not secure 20.163.9.187:3000/dashboard

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Verify it's you

⋮

Care Call

DashboardReportsSettings

Logout🌙

Active Calls

Room 4

10022 mins ago

Unmuted

Mute

Total Calls

1

📞

Pending Calls

0

😊

Resolved Today

0

✓

Response Time

-

🕒

Recent Activity

View all

Quick Actions

REPORT PAGE:

Search by Room or Call ID

All Statuses

All

All Rooms

Start Date: dd-mm-yyyy

to

End Date: dd-mm-yyyy

Export Excel

Export PDF

Summary Reports

Calls per DayCalls per RoomAttending Lag

Call History

ROOM	DEPARTMENT	FLOOR	STATUS	MUTED	CREATED	MUTED AT	RESET AT	REPEAT COUNT	LAST REPEAT AT
Room 11	General Ward	2	Resolved	Unmuted	5/19/2026, 2:40:34 PM		5/19/2026, 3:03:39 PM	0	
Room 10	Intensive Care	2	Active	Unmuted	5/19/2026, 2:38:18 PM			0	

SETTINGS:

← → ↻ ⚠ Not secure 20.163.9.187:3000/settings 🔍 ☆ 📄 Verify it's you ⋮

Care Call

DashboardReportsSettings

Logout🌙

Manage Rooms

Rooms

+ Add New Room

Room Name	Device No	Room Type	Department	Floor	Actions
Room 1	11	Emergency	General Ward	2	Edit Delete
Room 2	11333	General	Emergency	2	Edit Delete
Room 3	113344	Private	Surgery	2	Edit Delete
Room 4	12	General	General Ward	2	Edit Delete
Room 5	101	Private	Intensive Care	2	Edit Delete
Room 6	208	ICU	Emergency	2	Edit Delete

Device Emulator:

Device Emulator

Org ID

ORG001

Floor Number

2

Room Number

200

Call (status -1)

Reset (status 0)

Success: Call status reset

N

Conclusion:

The **Nurses Calling System** helps improve communication between patients and healthcare staff. It enhances hospital efficiency and ensures timely medical assistance. Implementing such systems leads to better patient satisfaction and improved healthcare service.